



Research Methodology

APUMP2408AI(PR)-ODL

Context-Aware Nepali Sign Language Translation using Hand
Sequences and Facial Cues

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Abstract

Nepali Sign Language (NSL) is a natural visual language used by Deaf communities in Nepal, but computational research on NSL remains in an early stage compared with better-resourced sign languages. The main constraint is not only the scarcity of public datasets but also the absence of continuous, sentence-level resources that would support translation rather than isolated sign classification. Recent literature on sign language processing shows that sign languages are fundamentally multimodal: meaning is encoded through manual features such as handshape and motion, and non-manual features such as facial expression, gaze, head movement, and body posture. These non-manual markers are not decorative; they contribute grammatical and semantic information, especially in questions, negation, emphasis, and discourse structure. This literature review synthesizes recent journal research on NSL resources, continuous sign language recognition, sign language translation, non-manual marker modeling, multimodal fusion, and context-aware evaluation. The review argues that NSL research is still at a foundational stage because the field lacks a public continuous NSL corpus and a translation architecture that explicitly models face and context alongside the hands. Based on current evidence, the most promising direction is a multimodal, context-aware NSL pipeline built on continuous data collection, explicit non-manual annotation, and efficient sequence models.

Keywords: Nepali Sign Language (NSL), Sign Language Translation (SLT), Sign Language Recognition (SLR), Continuous Sign Language Recognition (CSLR), Non-Manual Markers (NMMs), Multimodal Learning, Sign Language Machine Translation (SLMT), Long Short-Term Memory (LSTM), Bilingual Evaluation Understudy (BLEU), Recall-Oriented Understudy for Gisting Evaluation (ROUGE)

Introduction

Sign languages are fully developed human languages rather than simplified versions of spoken languages. They rely on multiple simultaneous channels of communication, including hand gestures, facial expressions, head movements, gaze, and torso posture. This makes sign language processing a difficult multimodal problem because a model must learn not only visual appearance but also the grammatical role of non-manual cues and the timing relationships among them. Recent surveys emphasize that sign language translation and recognition are distinct but related tasks, and that the field has moved from simple isolated classification toward broader problems such as translation, production, and discourse-aware interpretation (Liang et al., 2023; Rastgoo et al., 2024; Violet & Sri, 2025).

For NSL, computational research is still limited. The NSL23 dataset is an important milestone because it gives researchers a publicly described benchmark for Nepali manual alphabet recognition, but it is still an isolated dataset rather than a continuous corpus. That means it can support recognition experiments, yet it cannot by itself support translation of natural signed sentences. The broader sign language literature shows that data scarcity is one of the main bottlenecks in moving from recognition to translation, especially in low-resource settings where continuous video, aligned text, and linguistic annotations are not yet available (Sunuwar et al., 2024; Alaghband et al., 2023; Núñez-Marcos et al., 2023).

This literature review is organized around one central question: what would it take to build a context-aware NSL translation system? The answer, according to recent research, is not simply to use a better classifier. Instead, a viable system will require continuous signing data, explicit modeling of non-manual markers, careful multimodal fusion, and evaluation methods that reward linguistic completeness rather than surface-level similarity alone. That conclusion is consistent with recent surveys of sign language recognition, translation, and production, as well as with empirical studies that show facial expression and other non-manual cues improve translation quality (Zheng et al., 2021; Gu et al., 2024).

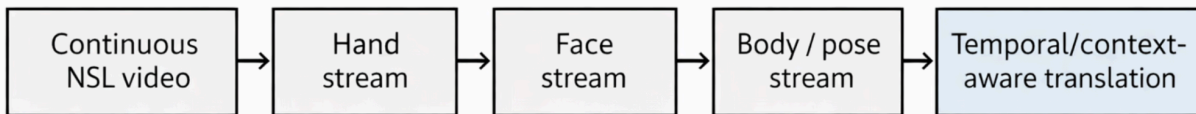


Figure 1. Proposed context-aware NSL translation pipeline.

The Landscape of Nepali Sign Language Research

The computational literature on NSL remains small. The most visible recent contribution is NSL23, which provides a dataset for Nepali sign language alphabets and demonstrates that deep learning can perform well on isolated recognition in a constrained setting. The dataset is valuable because it establishes a baseline for future work, but it is limited in scope: alphabet-level classification is not the same as sentence-level interpretation, and isolated signs do not capture the temporal and grammatical complexity of real communication. For a translation system, this distinction matters greatly because the target is not a label for a single handshape but a sequence of signs embedded in context (Sunuwar et al., 2024).

The broader sign language research literature consistently shows that dataset availability determines what kinds of models can be built. Surveys by Rastgoo et al. (2021), Alaghband et al. (2023), Núñez-Marcos et al. (2023), and Rastgoo et al. (2024) all identify the lack of

high-quality datasets as a core obstacle to progress. They also emphasize that while isolated recognition has advanced rapidly, continuous recognition and translation lag behind because they require aligned video-text corpora, sign boundary information, and often gloss annotations. This is especially true when one wants to build systems that generalize across signers, camera conditions, and natural signing speed.

NSL should be viewed in that same low-resource context. The literature reviewed here does not reveal a public continuous NSL corpus comparable to large resources in other sign languages. That absence is important because continuous corpora enable temporal modeling, contextual interpretation, and multimodal annotation. In contrast, a dataset limited to isolated signs can support recognition experiments, but cannot train a translation system that must learn how signs connect into clauses and sentences. This gap is the main reason NSL translation remains underdeveloped in the computational literature.

While Nepali Sign Language (NSL) is computationally low-resource, its structural and historical relationship to Indian Sign Language (ISL) offers a viable pathway for cross-lingual transfer learning. Comparative studies by Woodward (1993) establish that sign languages in India and Nepal belong to the same South Asian sign language family. Because they share deep grammatical and lexical overlaps, pre-training sequential models on large-scale ISL datasets provides a statistically heavy visual baseline. This aligns with modern architectural surveys (Toshpulatov et al., 2025; Rastgoo et al., 2024), which show that transfer learning and multimodal sequence modeling are standard practices for low-resource sign settings. This baseline can then be fine-tuned on smaller, localized continuous NSL datasets, maintaining accuracy while bypassing extreme data shortages.

A useful comparison is How2Sign, which demonstrated how continuous corpora can change the trajectory of sign language research by enabling multimodal, sentence-level modeling. Once structured continuous data exists, researchers can move from simple sign classification toward end-to-end translation, co-articulation modeling, and context-aware decoding. For NSL, the immediate research need is therefore not merely better algorithms but a corpus that captures natural signing in a form suitable for sequence modeling.

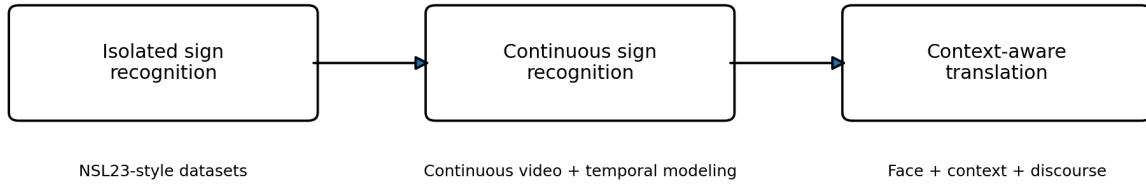


Figure 1. Research progression for Nepali Sign Language: from isolated recognition to context-aware translation.

Why Isolated Recognition Is Not Enough

A recurring theme in the literature is the gap between isolated sign recognition and full translation. Isolated recognition treats each sign as a separate class, which is useful for benchmarks and educational tools, but real communication does not occur as a sequence of neatly separated labels. Continuous signing introduces co-articulation, variable speed, sign overlap, and ambiguity at sign boundaries. As a result, a model that performs well on isolated classification can still fail badly in sentence-level interpretation. Recent reviews stress that continuous sign language recognition and translation require a much richer modeling strategy than isolated recognition.

The literature also shows that sign language translation is not a simple word-for-word mapping. In spoken-language translation, sequence-to-sequence models can often rely heavily on text statistics. In sign language translation, however, the input is visual, temporal, and highly multimodal. The mapping from sign to text may involve glosses, intermediate representations, or direct end-to-end architectures. Surveys and recent modeling papers emphasize that this process is complicated by linguistic reordering, signer variation, and the fact that sign languages do not align neatly with the word order of spoken languages (Liang et al., 2023; Núñez-Marcos et al., 2023).

This is where the NSL problem becomes more demanding. If NSL translation is attempted using only isolated sign recognition, the resulting system may label hands correctly but still miss the utterance’s grammatical force. The same manual content can convey different meanings depending on facial expression or head movement. That means translation quality depends on correctly interpreting non-manual markers, not only on identifying individual signs. Recent

studies on facial information in neural sign language translation confirm this point by showing that facial expression can contribute substantially to translation accuracy (Zheng et al., 2021).

From a research-design perspective, this implies that NSL systems should not be evaluated only as isolated classifiers. A true NSL translation system must be judged on sentence-level correctness, handling of temporal flow, and preservation of grammatical meaning. The recent literature increasingly supports this view by calling for richer metrics and more realistic task definitions, especially when the goal is accessibility rather than laboratory benchmark performance.

Continuous Sign Language Recognition and Translation

Continuous sign language recognition has advanced significantly in the last few years. Recent surveys show a strong shift away from purely handcrafted pipelines and toward deep learning systems that combine visual feature extraction with temporal modeling. Earlier methods often relied on frame-wise features or classical sequence models, but recent work increasingly uses transformer architectures, self-attention, and multimodal fusion to capture longer temporal dependencies. This is important because continuous signing is not just a series of static poses; it is a flowing signal in which the meaning of one segment depends on neighboring segments.

The recent review literature also shows that sign language translation should be understood as part of a broader sign language processing ecosystem that includes recognition, translation, and production. Rastgoo et al. (2024) argue that sign language production must deal with challenges such as dataset scarcity, high-resolution output generation, and the need to model realistic signing motion. That observation is relevant to translation because a bidirectional system often requires both interpretation of incoming signs and generation of linguistically appropriate output. NSL translation should therefore be designed with the understanding that recognition and production are coupled research problems.

Recent empirical papers show how this shift is happening in practice. Gu et al. (2024) demonstrate sign language recognition and translation using wearable inertial motion capture, showing that multimodal sensing can support end-to-end sentence recognition. Uddin et al. (2025) similarly show that MediaPipe-based features combined with LSTM temporal modeling can achieve strong results for Norwegian Sign Language, and they note that hand and mouth landmarks help distinguish similar signs. These papers are important for NSL because they show that temporal modeling and modality integration can be effective even in underrepresented sign languages.

The newer journal literature also points to more efficient architectures. Damdoo and Kumar (2025) explicitly target edge deployment, addressing memory constraints in sign language translation. That is highly relevant for a low-resource language such as NSL because a future translator may need to run on modest hardware in schools, community centers, or mobile platforms. A context-aware system for NSL does not have to be the largest possible model; it has to be the most feasible model that still preserves linguistic detail.

The Critical Role of Non-Manual Markers

A major finding across the literature is that non-manual markers are linguistically central. Facial expressions, eye gaze, head nods, head shakes, mouth patterns, and torso movement can mark questions, negation, topicalization, emphasis, and discourse structure. Zheng et al. (2021) directly show that ignoring facial expression information leads to translation errors in neural sign language translation. Their work is highly relevant because it demonstrates that non-manual features are not simply decorative or emotional; they can carry critical communication content.

More recent work continues the same line of reasoning. Liang et al. (2023) describe sign language translation as inherently multimodal, while Rastgoo et al. (2024) note that visual and language modalities are central to sign language production and translation. Gu et al. (2024) and Uddin et al. (2025) also show that hand motion alone is often insufficient, and that adding other visual channels can improve recognition and translation. Taken together, these studies make one point especially clear: a sign language model that ignores the face is not modeling the full language.

The annotation side of the problem is equally important. Chemnad and Othman (2025) review how avatar technologies depend on accurate monitoring of sign language acquisition and perception, which implies that a faithful computational model needs more than hand tracking. Shahin and Ismail (2025) similarly frame trustworthy sign language translation as a system-level problem that must preserve privacy, efficiency, and practical usability. In both cases, the implicit assumption is that non-manual cues are part of the system’s linguistic responsibility, not optional extras.

This matters especially for NSL because a future corpus should not be limited to manual signs. If only handshape and motion are recorded, the dataset will be too weak for context-aware translation. A question may be mistaken for a statement, negation may be lost, and emphasis may be erased. The literature on facial expression modeling, multi-channel translation, and

continuous recognition all suggests that NSL requires explicit annotation of non-manual markers from the outset. That is a design choice, not a later enhancement.

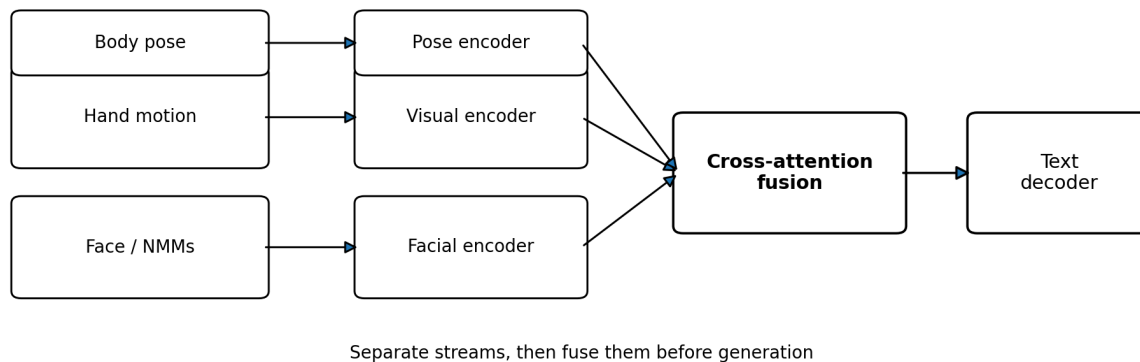


Figure 2. Suggested multimodal architecture for a context-aware NSL translation system.

Multimodal Fusion Strategies and Model Design

Because sign language is multimodal, fusion strategy is central to system design. Recent literature shows that useful models often combine multiple streams rather than compressing everything into a single visual representation. Sadeghzadeh et al. (2024), for example, propose a multilingual multimodal sign language recognition system that combines hand-crafted and deep-learning features for stronger robustness. The key lesson is that multimodal inputs can improve recognition when they are designed to complement one another rather than simply being concatenated.

Context-aware translation is another important fusion strategy. Lepp et al. (2025) argue for co-creation in sign language processing and translation technology, which is particularly important because system design should reflect the needs of Deaf communities and not only the preferences of engineers. In practice, that means NSL dataset design, annotation rules, and evaluation criteria should be created with fluent signers and linguists. The same paper also implies that meaningful translation systems should be developed with user-centered design in mind.

The evaluation literature also shapes fusion design. Traditional sign language translation metrics can easily undercount non-manual meaning when they reduce output to a single linear sequence.

The journal literature therefore increasingly stresses that evaluation should consider linguistic fidelity, non-manual preservation, deployment practicality, and trustworthiness together (Liang et al., 2023; Shahin & Ismail, 2025; Chemnad & Othman, 2025). For NSL, this means the success of a model should be judged not only by surface overlap with reference text but also by whether it correctly preserves questions, negation, emphasis, and discourse role.

A realistic NSL architecture, then, would likely include at least three components: a visual stream for hand and body motion, a facial stream for non-manual markers, and a language modeling or contextual decoding component that can incorporate discourse cues. Such a model need not be fully end-to-end at the start. In a low-resource setting, a staged approach may be more practical: first pretrain visual encoders on unlabeled data, then fine-tune on a smaller annotated corpus, and finally add context-aware decoding once sentence-level translation data is available. This strategy aligns well with the direction suggested by Damdoo and Kumar (2025), Shahin and Ismail (2025), and Violet and Sri (2025).

Challenges and Gaps for NSL Translation

The main gap for NSL is still the absence of a continuous corpus that can support sentence-level translation. The NSL23 dataset is valuable, but it is not enough. A proper corpus would need fluent continuous signing, multiple signers, varied contexts, and aligned annotation of both manual and non-manual elements. Without such a dataset, models can only be approximated by analogy with other sign languages, which limits reliability and makes evaluation difficult.

A second gap is annotation quality. Non-manual markers are difficult to annotate consistently, and recent work on sign language perception and monitoring shows that user variability affects how such data should be modeled. That means NSL corpus creation will require clear annotation guidelines, quality control, and likely some semi-automated tools to reduce manual burden. The challenge is not just collecting video but making sure the corpus captures the grammatical cues needed for translation.

A third gap is evaluation. Traditional metrics such as BLEU, ROUGE, or word error rate are useful, but they do not fully capture whether a system preserved non-manual meaning, discourse relations, or contextual force. The journal literature increasingly argues for evaluation practices that better reflect the structure of sign languages and the practical needs of Deaf users (Liang et al., 2023; Shahin & Ismail, 2025; Chemnad & Othman, 2025). For NSL, a high automatic score will not be enough if the system still confuses questions with statements or drops the facial information that gives the utterance its grammatical shape.

A fourth gap is computational feasibility. Deep transformers and multimodal pipelines can be expensive to train and deploy. However, recent journal papers show that efficiency-aware methods are possible. SignEdgeLVM targets edge deployment, and several review papers stress that future systems must be both accurate and practical. This is important for NSL because a translation tool that only works in a research lab will have limited social value. The final system should be robust enough for real users, real cameras, and real constraints.

A fifth gap is community involvement. Lepp et al. (2025) make a strong case for co-creation in sign language processing and machine translation technology. For NSL, that means signers should participate in deciding what data is collected, how it is labeled, and what counts as a useful translation. This is not only an ethical point but also a technical one, because model performance depends on whether the dataset reflects real communication. A context-aware NSL project should therefore be designed as a collaborative linguistic and engineering effort.

Future Research Direction for Nepali Sign Language

The reviewed literature supports a clear future direction. First, NSL needs a continuous corpus rather than only isolated alphabet data. That corpus should include fluent signers, sentence-level signing, and sufficient variation in age, signing style, and recording conditions. Second, the corpus should annotate non-manual markers explicitly, especially facial expression, head movement, gaze, and mouth patterns. Third, the modeling pipeline should be multimodal, with separate streams for hand motion, pose, and facial features, followed by a context-aware decoder. These steps are consistent with the latest work on continuous recognition, multimodal fusion, and sign language translation.

A practical NSL system would probably benefit from a staged research plan. In the first stage, unlabeled or weakly labeled video could be used to pretrain visual encoders. In the second stage, a smaller annotated corpus could fine-tune recognition and translation components. In the third stage, the system could be evaluated with both automatic metrics and fluent signers' judgments. This kind of development path is in line with the broader field's movement toward self-supervised learning, efficient architectures, trustworthy design, and co-created deployment (Damdoo & Kumar, 2025; Lepp et al., 2025; Shahin & Ismail, 2025).

Another important direction is context. The recent translation literature shows that sign language meaning often depends on more than the current sentence. Context-aware translation can incorporate previous utterances, scene information, or discourse history, improving accuracy when visual signals are ambiguous. For NSL, this is especially promising because it aligns with

the real-world way people communicate: meaning is built over time, not one isolated sign at a time. A future NSL translator should therefore be designed to use local visual features and broader discourse cues together.

In summary, the strongest evidence suggests that NSL research should move away from isolated-sign benchmarks and toward a corpus-driven, multimodal, and community-informed translation framework. The literature already provides the key ingredients: dataset design principles, evidence for non-manual markers, efficient temporal modeling, and calls for co-created, trustworthy systems. The missing piece is a continuous NSL resource that brings these ideas together. Building that resource would not only advance research but also create a practical bridge between Deaf and hearing communities in Nepal.

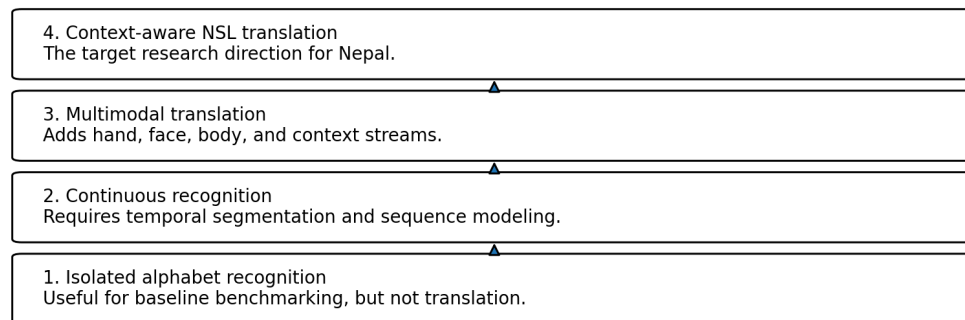


Figure 3. Research maturity gap in NSL, moving from baseline recognition toward a useful translation system.

Synthesis of Key Journal Studies

Across the last five years, the literature has developed in a fairly consistent direction. Early survey articles established that sign language recognition had already benefited from deep learning, but also noted persistent problems in signer variation, dataset scarcity, and the limited handling of non-manual cues. Rastgoo et al. (2021) provide a broad taxonomy of isolated and continuous recognition, which is useful because it clarifies that these tasks have very different data and modeling requirements. Alaghband et al. (2023) extend that argument by showing that the field still lacks robust datasets and software tools, while Núñez-Marcos et al. (2023) make

clear that machine translation is even more difficult than recognition because it must preserve the meaning structure of the source language rather than merely classify it.

The more recent production-oriented literature takes the analysis further. Rastgoo et al. (2024) argue that sign language production is a necessary complement to recognition and translation, because a full sign language interface should ideally operate in both directions. That perspective is important for NSL because it implies that a future system should not be designed as a one-way recognizer alone. Instead, it should be viewed as part of a bidirectional accessibility stack that can support recognition, translation, and possibly avatar-based generation. This aligns with Shahin and Ismail (2025), who frame trustworthy SLMT as a deployment problem as well as a modeling problem.

Several empirical papers support the importance of multimodal input. Zheng et al. (2021) show that facial expression contributes directly to translation quality, while Gu et al. (2024) demonstrate that wearable inertial data can support sentence-level ASL recognition and translation. Uddin et al. (2025) add evidence from Norwegian Sign Language showing that even lightweight landmark-based systems can be useful when the data representation matches the signing structure. Sadeghzadeh et al. (2024) further show that illumination-invariant multimodal recognition can improve robustness, which matters for real-world data where lighting, camera quality, and signer distance vary.

Another important strand is efficiency and deployment. Damdoo and Kumar (2025) propose a transformer model designed for edge devices, making the point that sign language translation systems do not have to be computationally excessive to be useful. This is important for low-resource languages like NSL because practical deployment in schools, clinics, and community centers may require modest hardware. Lepp et al. (2025) strengthen this argument from a human-centered angle by emphasizing co-creation. A translator that is technically sophisticated but disconnected from user needs will not be effective. For NSL, co-design with fluent signers would improve both the linguistic quality of the corpus and the social acceptability of the eventual system.

The evaluation literature points in the same direction. Liang et al. (2023) and other recent journal studies show that sign language translation should be measured in ways that preserve linguistic fidelity and do not collapse all multimodal information into a simplified sequence. Chemnad and Othman (2025) complement this by showing that avatar and perception research depends on careful monitoring of how people acquire and understand sign language. In other words,

evaluation is not just a technical metric; it is part of how the system remains faithful to sign language itself.

These studies collectively suggest a research roadmap for NSL. The first phase should prioritize dataset construction and annotation standards. The second phase should use multimodal pretraining and lightweight temporal modeling. The third phase should incorporate context-aware translation and multi-channel evaluation. Figure 1 in this report summarizes that progression visually, moving from isolated sign recognition toward context-aware translation. Figure 2 illustrates the kind of multimodal architecture that is most appropriate for the language, while Figure 3 shows the maturity gap that still separates current NSL research from a fully usable translator.

Overall, the literature does not support a narrow hand-only solution. It supports a layered strategy in which the hand stream carries lexical content, the facial stream captures grammatical and pragmatic cues, the pose stream captures body dynamics, and the decoder integrates context into fluent text output. That layered strategy is not merely an engineering choice. It is the best match to the linguistic reality of sign languages and therefore the most appropriate direction for future NSL research.

Conclusion

This literature review shows that NSL translation is still at a foundational stage. The publication of NSL23 is an important starting point, but the field still lacks the core resource needed for translation: a public continuous corpus with aligned linguistic annotation. At the same time, the broader sign language literature makes the research path fairly clear. Recent papers show that sign languages are multimodal, that facial and other non-manual markers are grammatically essential, and that continuous translation requires temporal modeling, multimodal fusion, and careful evaluation.

The evidence also suggests that a successful NSL translation system should not be built as a hand-only classifier with a text output layer attached. It should instead be developed as a context-aware multimodal system that models hands, face, body posture, and discourse context together. Recent journal articles on continuous recognition, edge-aware translation, co-creation, and trustworthy design show that such a system is both technically plausible and socially necessary. The next step for NSL research is therefore not just better accuracy; it is better representation of the language itself.

A final practical point is that the figures in this report are intentionally simple and original. They are meant to clarify the research logic rather than replace empirical evidence. Figure 1 shows the progression from isolated recognition to continuous recognition and then to context-aware translation. Figure 2 shows why a multi-stream architecture is a better fit for NSL than a hand-only model. Figure 3 summarizes the maturity gap that still separates the current NSL literature from the type of system that would be useful in real communication settings. Together, they reinforce the main conclusion of this review: the next NSL milestone should be corpus creation, not just model tuning.

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Literature Review Matrix

The table below summarizes the core journal sources used in the review.

Author / article title / journal	Research question(s)	Methodology	Analysis and results	Conclusions	Implications for future research / practice
Sunuwar et al. (2024) NSL23 dataset for alphabets of Nepali sign language Data in Brief, 53, 110080	Benchmark dataset for Nepali alphabet recognition?	Dataset creation + transfer-learning classification	630 clips / 1205 gestures; strong isolated-recognition baseline.	NSL now has a foundational dataset, but only for isolated signs.	Extend to continuous signing and sentence-level annotation.
Liang et al. (2023) Sign language translation: A survey of approaches and techniques Electronics, 12(12), 2678	What approaches dominate sign language translation?	Systematic survey	Sign-to-gloss-to-text and gloss-free methods are both important.	Translation is multimodal and context-sensitive.	NSL should adopt multimodal, context-aware design.
Núñez-Marco s et al. (2023) A survey on sign language machine	What are the main SLT technical challenges?	Survey of SLT methods	Data scarcity, alignment, and evaluation	High-quality corpora are the main bottleneck.	NSL needs continuous data and robust evaluation.

translation Expert Systems with Applications, 213, 118993			are recurring issues.		
Zheng et al. (2021) Enhancing neural sign language translation by highlighting facial expression information Neurocomputing, 464, 462–472	Do facial expressions improve translation?	Neural translation model with facial emphasis	Facial cues improved translation quality.	Non-manual markers are linguistically important.	NSL models should include a face stream.
Rastgoo et al. (2021) Sign language recognition: A deep survey Expert Systems with Applications, 164, 113794	What are the major directions in SLR?	Deep survey	The field moved from handcrafted features to deep learning.	Recognition is mature but not equal to translation.	NSL should move beyond isolated classification.
El-Alfy & Luqman (2022) A comprehensive survey and taxonomy of sign language research Engineering Applications of AI, 114, 105198	How should sign language research be organized?	Taxonomy and survey	Organizes work across recognition, translation, and production.	Task definitions matter.	NSL research should choose tasks carefully.
Alaghband et al. (2023) A survey on sign language	What are the recent trends in the field?	Broad survey	Highlights data shortage and	Existing datasets are too narrow.	NSL needs a richer corpus.

literature Machine Learning with Applications, 14, 100504			modality-fusion trends.		
Gu et al. (2024) ASL recognition and translation using wearable inertial motion capture system Sensors, 24(2), 453	Can wearable sensing support translation?	Wearable inertial capture + translation pipeline	Multimodal sensing improved recognition and translation potential.	Alternative sensing can complement video.	NSL may benefit from pose/motion sensing.
Sadeghzadeh et al. (2024) MLMSign: Multilingual multimodal illumination- invariant sign language recognition Intelligent Systems with Applications, 22, 200384	Can multimodal fusion improve robustness?	Multilingual multimodal recognition	Fusion improves robustness under lighting changes.	Fusion helps with real-world variability.	NSL datasets should capture varied environments.
Rastgoo et al. (2024) A survey on recent advances in sign language production Expert Systems with Applications, 243, 122846	What changed in sign language production research?	Survey of production models	Production faces realism, data, and multimodality challenges.	Production and translation are linked.	NSL should consider both translation and production.

Lepp et al. (2025) Co-creation for sign language processing and translation technology Information, 16(4), 290	Why is co-creation important?	Participatory review	Community involvement improves relevance and trust.	User-centered design is essential.	NSL should be built with signer participation.
Chemnad & Othman (2025) Perception and monitoring of sign language acquisition for avatar technologies Multimodal Technologies and Interaction, 9(8), 82	How do users perceive avatar-based sign technologies?	Rapid focused review	Highlights monitoring and perception considerations.	Avatar systems must reflect real linguistic behavior.	NSL evaluation should include usability.
Damdoo & Kumar (2025) SignEdgeLV M transformer model for enhanced sign language translation on edge devices Discover Computing, 28(1), 15	Can translation be made efficient for edge devices?	Edge-focused transformer design	Improves feasibility on low-resource hardware.	Efficiency is critical for deployment.	NSL tools should be practical to deploy.
Gan et al. (2025) Vision-based sign language translation via a	Does skeleton information improve translation?	Skeleton-aware neural network	Pose features strengthen motion representation.	Body posture is useful beyond RGB video.	NSL should include pose-based features.

skeleton-aware neural network Journal of Computer Science and Technology, 40(2), 378–396					
Toshpulatov et al. (2025) Deep learning pathways for automatic sign language processing Pattern Recognition, 164, 111475	What are the major deep-learning pathways?	Systematic review	The field is moving toward multimodal sequence models.	End-to-end pipelines are increasingly common.	NSL should follow the same multimodal trajectory.